IIT ROORKEE

NPTEL ONLINE CERTIFICATION COURSE

Charging Infrastructure

Lecture-59

Charging Stations-II

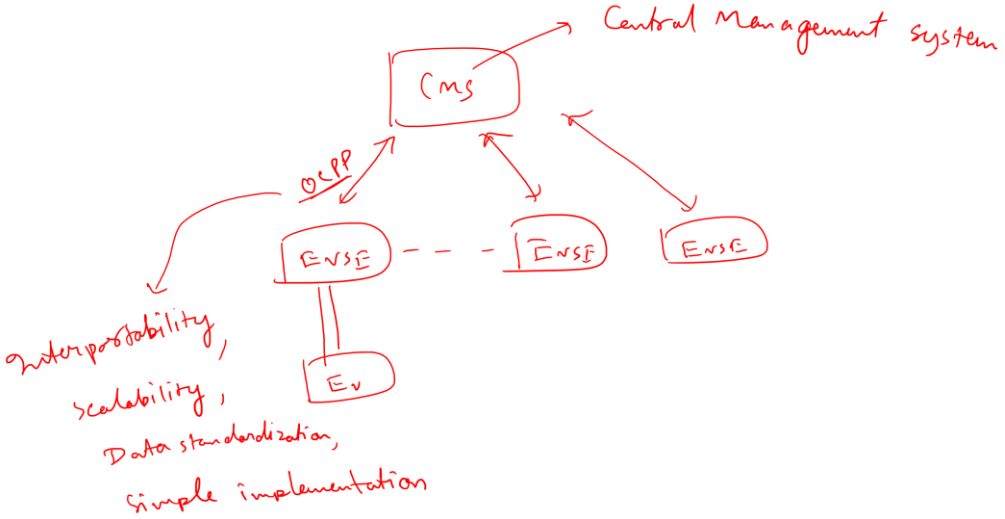
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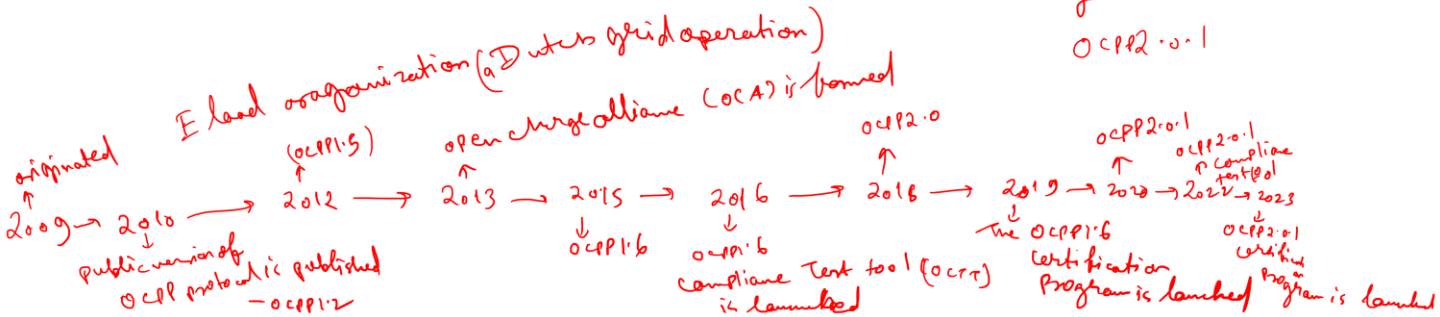
Recap

- Charging stations
- OCPP protocol



Recap: OCPP

- OCPP provides a standardized protocol for communication between EVSE and central management systems. (CMS) or CSMS
- The protocol enables the exchange of messages and data between EVSE and central management systems.
- OCPP standard has some key components: OCPP profiles, data types, messages, configuration keys, use case and test case.
- OCPP profile: define how the two entities communicate, including the messages exchanged, the message format, and the specific functionality of each system. entity



Recap:OCPP

- Message: data unit transmitted between EVSE and CMS during a charging session. It has a specific format and purpose.
- Data type: OCPP profiles utilize particular data types to specify the format and organization of data including Boolean, string, date-time, enumeration, etc.
- Configuration keys: Configuration key specifies different settings and restrictions that manage how the EVSE operates throughout a charging session. This covers charging rates, maximum power limits, and authorization requirements. Every profile contains a collection of predefined configuration keys.
- Use case: A use case describes a particular situation that the protocol is intended to handle. Examples of use cases are start transaction, stop transaction, etc.
- Test cases: Developers rely on them to confirm the proper implementation of the OCPP protocol. Each OCPP profile contains a collection of predefined test cases that help developers validate the behavior of their implementation.

Source: https://www.ampeco.com/guides/complete-ocpp-guide/

OCPP Implementation

- There are two types of implementation for OCPP:
  - OCPP SOAP Implementation (for OCPP 1.6 and earlier versions): Simple Object ACESS Protocol (SOAP) is a messaging protocol for data exchange
    - It utilizes XML for data representation: large size and heavy duty due to protocol headers.
    - It requires both the charge point and the CMS to act as servers to facilitate two-way communication.
  - OCPP JSON Implementation (for OCPP 1.6 version and above): JavaScript Object Notation (JSON) is a data interchange format used to represent the data
    - It is lightweight and flexible format of data representation: human readable and compact.
    - It relies on a persistent WebSocket based connection for two-way communication.
- WebSocket is a communication protocol which allows real-time, two-way communication over a single always open connection.

Source: https://www.ampeco.com/guides/complete-ocpp-guide/
Open Charge Point Protocol document released by OCA (open charge alliance)

OCPP Version Differences

- OCPP 1.6 offers various functionalities, such as status notifications, remote start and stop of a charging session, firmware management, and data transfer via a WebSocket connection using JSON. Additionally, It support SOAP.
- OCPP 1.6 has some limitations, regarding its security features. It provides only fundamental smart charging capabilities and lacks support for the ISO 15118 protocol, which enables advanced EV-to-charger communication and functionalities such as Plug & Charge.
- OCPP 2.0.1 offers enhanced security features, incorporates advanced smart charging functionalities, supports the ISO 15118 standard, is highly scalable, and enables direct vehicle communication.

↓
implement complex
Communication
architecture

Source: https://www.ampeco.com/guides/complete-ocpp-guide/

OCPP Messages Format for OCPP-J Implementation

- Three different messaging types: CALL, CALLRESULT, and CALLERROR.
- Messages between the CMS and EVSE is a string of CALLs and CALLRESULTs.

MessageType	MessageTypeNumber
CALL	2
CALLRESULT	3
CALLERROR	4

Data taken from Open Charge Point Protocol JSON 1.6, OCPP-J 1.6 Specification document

OCPP-J Messages Format: CALL

Field	Datatype	Meaning
MessageTypeId	integer	This is a Message Type Number which is used to identify the type of the message.
MessageId	string[36]	This is a unique identifier that will be used to match request and result.
Action	string	The name of the remote procedure or action. This field will contain a case-sensitive string.
Payload	JSON	Payload is a JSON object containing the arguments relevant to the Action. If there is no payload JSON then notations: null or and empty object \{}.

- The message ID serves to identify a request. A message ID for a CALL message is different from all previously used message IDs.
- A message ID for a CALLRESULT or CALLERROR message MUST be equal to that of the CALL message

Data taken from Open Charge Point Protocol JSON 1.6, OCPP-J 1.6 Specification document

OCPP-J Messages Format: CALLRESULT

Field	Datatype	Meaning
MessageTypeId	integer	This is a Message Type Number which is used to identify the type of the message.
MessageId	string[36]	This must be the same ID which is in the call request. The recipient can match request and result.
Payload	JSON	Payload is a JSON object containing the arguments relevant to the Action. If there is no payload JSON then notations: null or and empty object \{}.

Data taken from Open Charge Point Protocol JSON 1.6, OCPP-J 1.6 Specification document

OCPP-J Messages Format: CALLERROR

Field	Datatype	Meaning
MessageTypeId	integer	This is a Message Type Number which is used to identify the type of the message.
MessageId	string[36]	This must be the same ID which is in the call request. The recipient can match request and result.
ErrorCode	string	This field must contain a string from the Error Codes table (given in the OCPP document).
ErrorDescription	string[255]	Should be filled in if possible, otherwise a clear empty string ("").
ErrorDetails	JSON	This JSON object describes error details in an undefined way. If there are no error details then fill in an empty object {}.

Data taken from Open Charge Point Protocol JSON 1.6, OCPP-J 1.6 Specification document

OCPP 1.6J Messages

- There are many messages: Authorize, BootNotification, CancelReservation, ChangeAvailability, Change Configuration, ClearCache, ClearChargingProfile, DataTransfer, DiagnosticsStatusNotifcation, FirmwareStatusNotifcation, GetConfiguration, GetDiagnostics, GetLocalListVersion, Heartbeat, MeterValues, RemoteStartTransaction, RemoteStopTransaction, etc.

- A total of 56 request and confirm messages are there.

Data taken from Open Charge Point Protocol JSON 1.6, OCPP-J 1.6 Specification document

OCPP 1.6J Message Payload Examples

Payload of RemoteStartTransaction.req

FIELD NAME	FIELD TYPE	DESCRIPTION
connectorId	integer	Optional. Connector number on which the transaction is starting. connectorId SHALL be > 0
idTag	IdToken	Required. The identifier that Charge Point must use to start a transaction.
chargingProfile	ChargingProfile	Optional. Charging Profile to be used by the Charge Point for the requested transaction. ChargingProfilePurpose MUST be set to TxProfile

Payload of Heartbeat.conf

FIELD NAME	FIELD TYPE	DESCRIPTION
current Time	dateTime	Required. This contains the current time of the Central System.

Data taken from Open Charge Point Protocol JSON 1.6, OCPP-J 1.6 Specification document

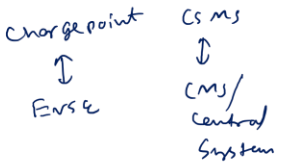
OCPP 1.6J Message Payload Examples

Payload of BootNotifcation.req

FIELD NAME	FIELD TYPE	DESCRIPTION
chargeBoxSerialNumber	CiString25Type	Optional. This contains a value that identifies the serial number of the Charge Box inside the Charge Point. Deprecated, will be removed in future version
chargePointModel	CiString20Type	Required. This contains a value that identifies the model of the ChargePoint.
chargePointSerialNumber	CiString25Type	Optional. This contains a value that identifies the serial number of the Charge Point.
chargePointVendor	CiString20Type	Required. This contains a value that identifies the vendor of the ChargePoint.
firmwareVersion	CiString50Type	Optional. This contains the firmware version of the Charge Point.
iccid	CiString20Type	Optional. This contains the ICCID of the modem's SIM card.
imsi	CiString20Type	Optional. This contains the IMSI of the modem's SIM card.
meterSerialNumber	CiString25Type	Optional. This contains the serial number of the main electrical meter of the Charge Point.
meterType	CiString25Type	Optional. This contains the type of the main electrical meter of the Charge Point.

Payload of BootNotifcation.conf

FIELD NAME	FIELD TYPE	DESCRIPTION
currentTime	dateTime	Required. This contains the Central System's current time.
interval	integer	Required. When RegistrationStatus is Accepted, this contains the heartbeat interval in seconds. If the Central System returns something other than Accepted, the value of the interval field indicates the minimum wait time before sending a next BootNotification request.
status	RegistrationStatus	Required. This contains whether the Charge Point has been registered within the Central System.



Data taken from Open Charge Point Protocol JSON 1.6, OCPP-J 1.6 Specification document

OCPP-J Messages Example

Boot Notification - Sent:

```
"messageTypeId": 2,
"uniqueId": "96197ee6-fb13-48fb-b0e0-
    d146a4187f83",
"action": "BootNotification",
"payload": {
  "chargingStation": {
    "model": "scorpio26",
    "vendorName": "shrishvesh"
  },
  "reason": "PowerUp"
}
```

Boot Notification - Response:

```
"messageTypeId": 3,
"uniqueId": "96197ee6-fb13-48fb-b0e0-
    d146a4187f83",
"payload": {
  "currentTime": "2025-04-29T12
     :23:48.683327+00:00",
  "interval": 10,
  "status": "Accepted"
}
```

Heartbeat - Sent:

```
"messageTypeId": 2,
"uniqueId": "ec136e00-e916-4f71-ab8b-
  b90edc375fe3",
"action": "Heartbeat",
"payload": {}
```

Heartbeat - Response:

```
"messageTypeId": 3,
"uniqueId": "ec136e00-e916-4f71-ab8b-
    b90edc375fe3",
"payload": {
  "currentTime": "2025-04-29T12:23:48Z"
}
```

Thank You